Data Sheet
Tuya Wi-Fi Water Level Sensor

Overview:

The Tuya Water Level Sensor is a 2.4Ghz Wi-Fi Sensor for water level monitoring. The sensor can measure a range from 10cm to 2.5m ± (1% Reading+10mm) at 20°C.

The sensor will upload data to Tuya Server and the user can access the water level via the Smart Life App, available on the app store.

The Sensor is suitable for water level monitoring in low or none condensing environment only.

Specifications:

Parameters	Specifications	Notes
Power Supply	DC 9V	
Current	Normal State <100mA	
Frequency	40kHz	
Operating Temp	-10°C - 70°C	
Communication	Wi-Fi (2.4GHz	
Measuring Range	10cm – 2.5m	
Blind Zone	10cm	
Measurement Tolerance	± (1% Reading+10mm)	20℃
Detection Cycle	≥200ms	
Detection Angle	Horizontal:1.7°（Typical Value）；Vertical Direction：12°~29°（Typical Value）	



Tuya Wi-Fi Water Level Sensor Installation Instructions

Important Note: Ensure Adequate Wi-Fi Coverage Before Sensor Installation

Prior to installing the sensor, it is crucial to confirm that there is ample Wi-Fi coverage in the chosen location. This ensures the sensor functions correctly and maintains a reliable connection.

1. **Placement:** Identify the optimal location for sensor placement based on the provided recommendations (page 2). Consider factors such as water level accuracy and Wi-Fi signal strength.

2. **Hole Preparation:** Drill a 72mm hole in the lid or top of the water tank using an appropriate drill bit or hole saw.

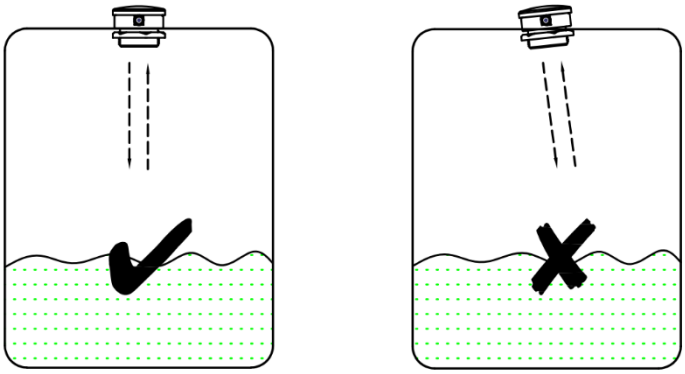
3. **Sensor Installation:** Securely screw the water level sensor into the prepared hole. Ensure a snug fit to prevent rain water ingress.

4. **Power Connection:** Connect the sensor to a power source following the provided power cable. Make sure the power connection is secure.

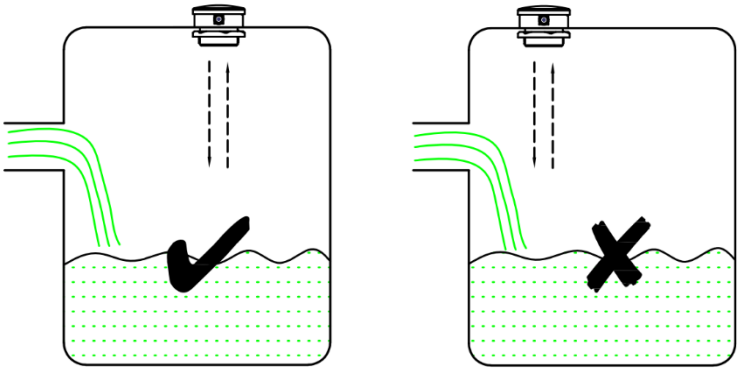
5. **Connection and Settings:** Refer to the accompanying leaflet for detailed instructions on connecting the sensor to your Wi-Fi network. Follow the step-by-step guide for configuring settings to ensure proper functionality.

Congratulations! Your Wi-Fi water level sensor is now installed and ready to monitor water levels effectively. For any additional assistance, refer to the comprehensive user manual included with the sensor or contact us.

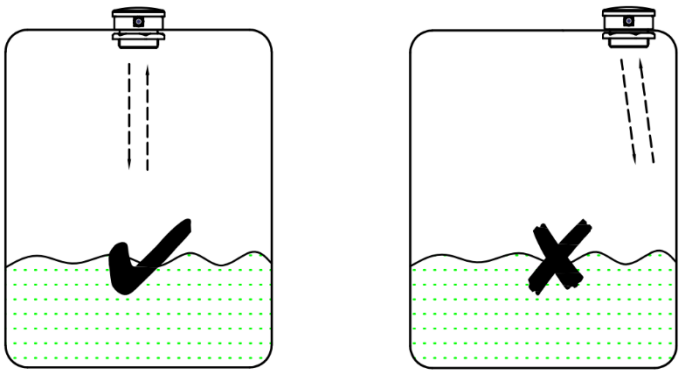
Recommended Sensor Placement



Sensor should be installed perpendicular to the liquid



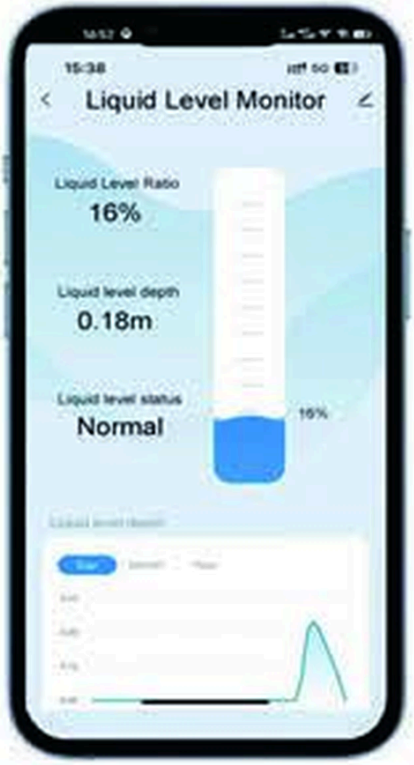
Install the sensor away from the water inlet to avoid pseudo-echo and false reading/notification



The sensor should not be installed close to the bulkhead, otherwise it will cause strong pseudo-echo and false readings

USER MANUAL




15:38
Liquid Level Monitor
Liquid Level Ratio
16%
Liquid level depth
0.18m
Liquid level status
Normal
16%

1. Overview

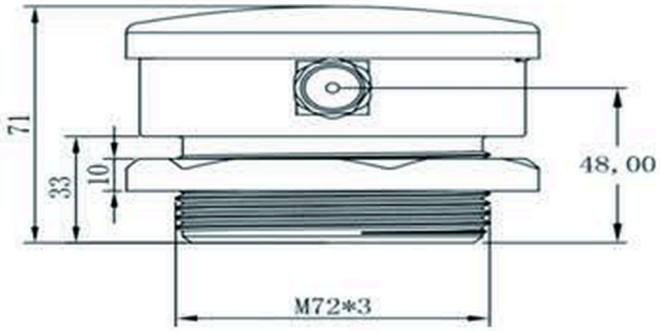
Thank you for purchasing the Wireless Ultrasonic Liquid Sensor with Tuya App Display. The ultrasonic level sensor measures the level without contact with the liquid. The sensor sends an ultrasonic signal, which is reflected back from the liquid surface. The fill level can thus be determined. The sensor can be used for oil and water tanks.

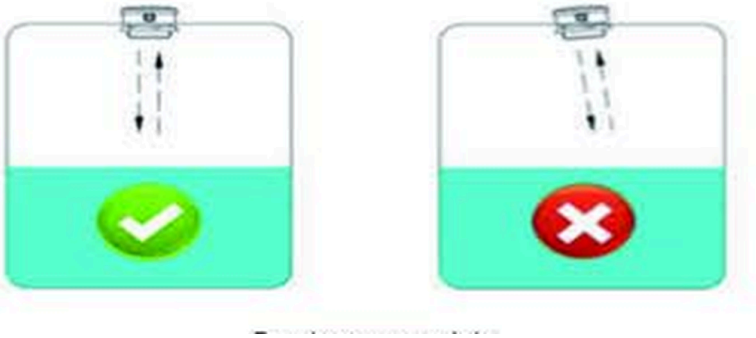


- Can be used for various liquids, since the ultrasonic sensor does not result in direct liquid contact
- Detect range: 10 cm to 2.9 m can measure high tanks
- High and low liquid alarm function
- Ideal for water/oil tanks
- Be able to monitor several tanks in one time.

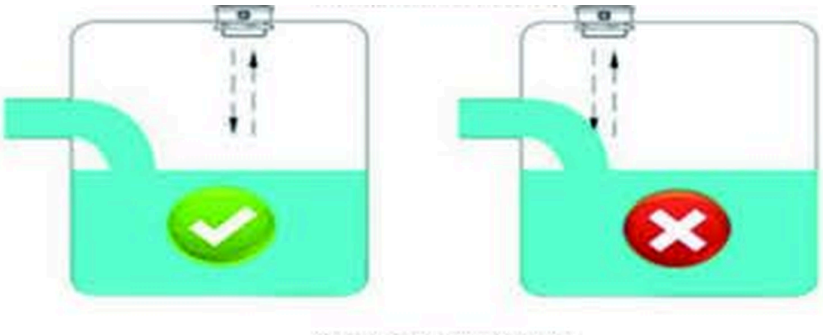
2. Mount the Sensor on the Tank

The sensor should be in the middle of the tank and not on a liquid inlet. This can falsify the measurement result. Please refer to the following pictures for the best installation positions. The hole should be 72 mm.

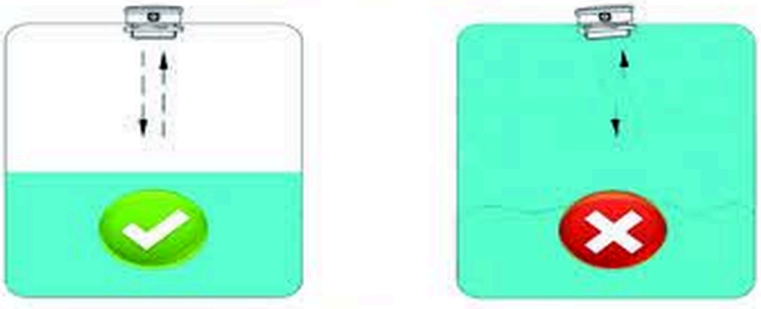




Put the sensor right



Away from tank wall



Sensor can’t touch liquid

3. Connection

1) Download "TUYA Smart APP" or "Smart Life APP" directly in the app store of mobile. Or scan the code below to download the app. iOS and Android devices are OK.



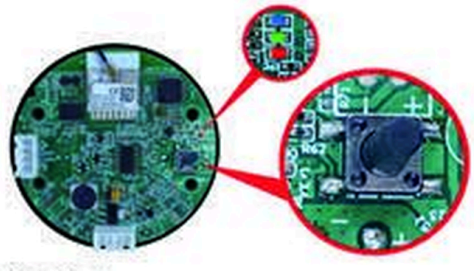


2) LED and button on sensor

- Red LED on/off: Power
- Green LED on/off: wifi connection indicator

Press the button for 5 s to connect wifi

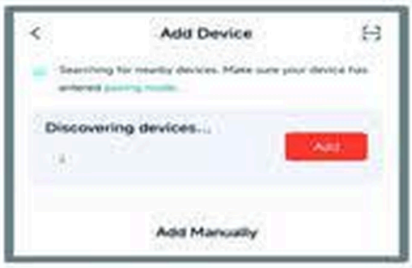
- Blue LED on/off: Mobile connection indicator



3) Turn on the Bluetooth on the mobile phone and open the TUYA Smart or Smart Life app.



Click Home – Add Device



Select Device

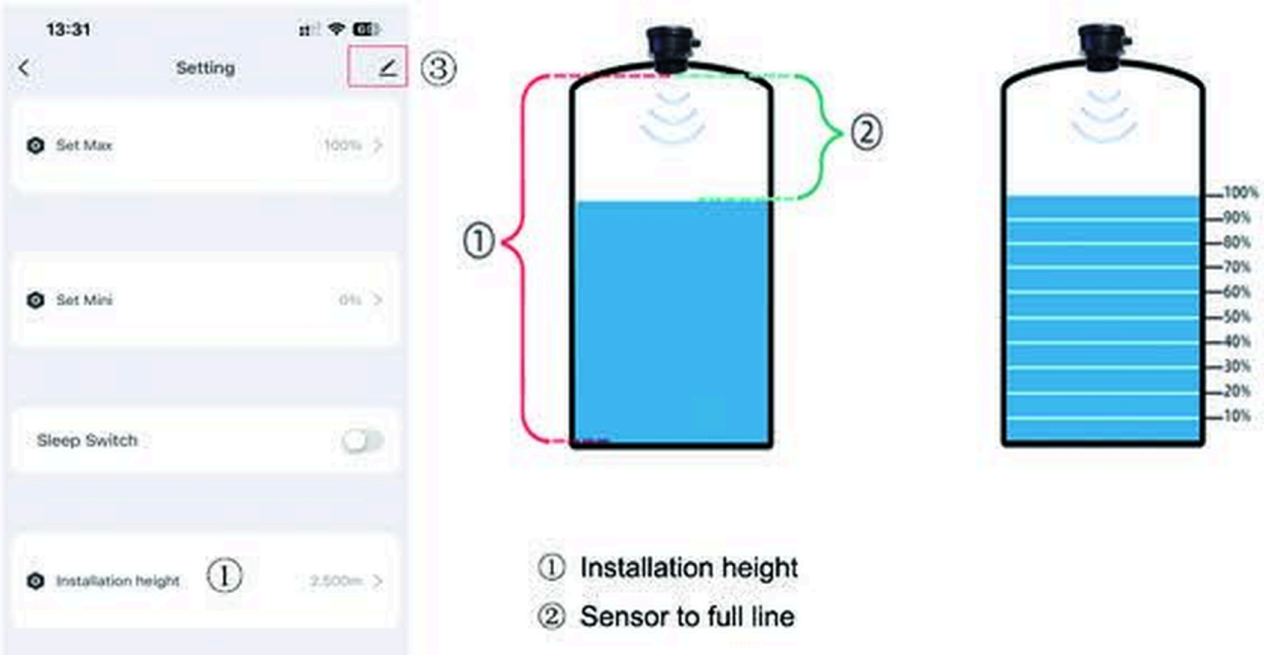


Enter WiFi account password

Note: WiFi frequency can only be 2.4 GHz, not 5 GHz.

When adding the device, make sure that the sensor and the mobile are connected to the same WiFi. After setting up, the sensor must work under wifi. If the mobile phone can't find the device or fails to connect the device, please click the PCB button twice to add the device again, or put the sensor close to the router and try again.

4. Setting



1) Set installation height ①

It is the distance from the sensor to the tank bottom.

Note: The installation height is different from the tank height because the sensor may be installed over the tank height.

The max installation height is 2.9 meters. If more than 2.9 meters, it is not suitable for this sensor.

2) Set sensor to liquid full line ②

It is the distance from the sensor to the full line.

Note: The blind zone is 0.1 meter. It can't be smaller than 0.1 meter.

3) Set Max for full level alarm

4) Set Min for low level alarm

5–7. Optional Function, Sharing & Alarm

5) Optional function

For Smart TLC2206-WF-S, it has a sleep mode which can save battery energy. In sleep mode, the sensor detects liquid level for 28 s for every 2 minutes.

6) Share to other device

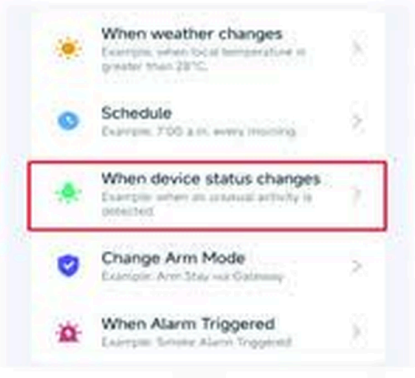
Click ③ to enter edit and choose Share Device.

7) Set alarm

Go back to the home page, click “Scenes”.



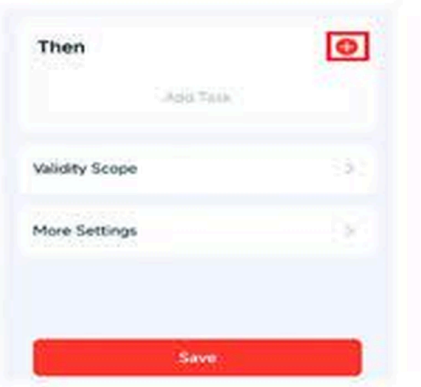
Create Scenes



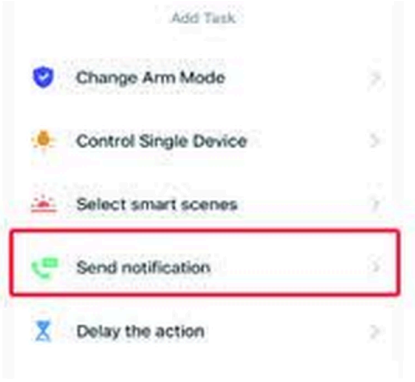
When device status changes



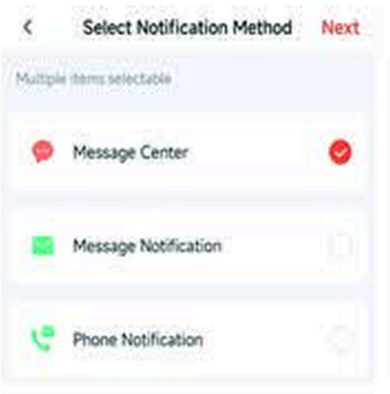
Liquid level status



Click “+”



Send notification



Message Center – Save

Edit the scene name, and click “Confirm”.